

CSE340: Computer Architecture

Assignment -2

Deadline: 14th March, 3:00 AM

Assignment must be handwritten. Scan and upload PDF in provided Google form

Question - 1:

Given a machine code, what do we understand by looking at the **Opcode** field?

Question - 2:

Suppose you have converted LD X9, 10(X21) to its equivalent RISC-V machine code. Now, based on which **field**, the machine will understand the size of the data being loaded into register X9?

Question - 3:

```
1. if (A[4] != B[7]){  
2.     if (A[4] == 0) {  
3.         A[4] = A[4] + 2;  
4.         sum = sum - 5  
5.     }else{  
6.         B[7] = B[7] / 16;  
7.     }  
8. }else{  
9.     B[7] = B[7] * 5  
10.    sum = sum + 10  
11. }
```

a) Construct the equivalent RISC-V code of the above-mentioned C code.

Base addresses of array A and B are in registers X20 and X21; Consider both arrays as double words. Sum is stored in register X22.

b) From your converted codes, pick any instruction from each of the three formats(R, I, S) and convert them into their corresponding binary values.

Note: For opcode, funct3 or any other field values that you do not know just put “X”.

Question - 4:

Given the following 32-bit binary machine codes, determine the corresponding RISC-V assembly instruction.

Use the provided opcode and funct reference table for identification

1. 00000000 01011 01010 000 01001 0110011
2. 00000000 00101 10001 101 10010 0010011

Instruction	Opcode	Funct3	Funct7
add	0110011	000	0000000
xor	0110011	100	0000000
ld	0000011	011	-
srli	0010011	101	0000000
sub	0110011	000	0100000